Factoring Polynomials

Sections

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Factoring Trinomials of the Form $ax^2 + bx + c$

Factoring Trinomials of the Form $x^2 + bx + c$ by

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Quadratic Equations and Problem

Solving

The Greatest Common Factor

Factors

Factors (either numbers or polynomials) When an

integer is written as a product of integers, each of the integers in the product is a ***factor*** of the original number.

When a polynomial is written as a product of polynomials, each of the polynomials in the product is a ***factor*** of the original polynomial.

Factoring – writing a polynomial as a product of polynomials.

Greatest Common Factor

Greatest common factor – largest quantity that is a factor of all the integers or polynomials involved.

Finding the GCF of a List of Integers or

Terms 1) Prime factor the numbers.

2) Identify common prime factors.

3) Take the product of all common prime factors. • If there are no common prime factors, GCF is 1.

Greatest Common Factor

Example

Find the GCF of each list of numbers.

1) 12 and 8

$12 = 2 \cdot 2 \cdot 3$

$8 = 2 \cdot 2 \cdot 2$

$8 = 2 \cdot 2 \cdot 2$

So the GCF is $2 \cdot 2 = 4$.

2) 7 and 20

$7 = 1 \cdot 7$

$20 = 2 \cdot 2 \cdot 5$ or (1, 20)

There are no common prime factors so the GCF is 1.

Greatest Common Factor

Example

Find the GCF of each list of numbers.

1) 6, 8 and 46

$6 = \mathbf{2} \cdot 3$

$8 = \mathbf{2} \cdot 2 \cdot 2$

$46 = \mathbf{2} \cdot 23$

$46 = \mathbf{2} \cdot 23$

So the GCF is 2.

2) 144, 256 and 300

$144 = \mathbf{2} \cdot \mathbf{2} \cdot 2 \cdot 3 \cdot 3$

$256 = \mathbf{2} \cdot \mathbf{2} \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

$300 = \mathbf{2} \cdot \mathbf{2} \cdot 3 \cdot 5 \cdot 5$

So the GCF is $2 \cdot 2 = 4$.

Greatest Common Factor

Example

Find the GCF of each list of terms.

1) x^3 and x^7

$x^3 = x \cdot x \cdot x$

$x = x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x$

$x^7 = x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x$

So the GCF is $x \cdot x \cdot x = x^3$

2) $6x^5$ and $4x^3$

$6x^5 = 2 \cdot 3 \cdot x \cdot x \cdot x \cdot x \cdot x$

$4x^3 = 2 \cdot 2 \cdot x \cdot x \cdot x$

So the GCF is $2 \cdot x \cdot x \cdot x = 2x^3$

Greatest Common Factor

Example

Find the GCF of the following list of terms.

a^3b^2, a^2b^5 and a^4b^7

$a^3b^2 = a \cdot a \cdot a \cdot b \cdot b$

$a\,b = a \cdot a \cdot a \cdot b \cdot b$

$a^2b^5 = a \cdot a \cdot b \cdot b \cdot b \cdot b \cdot b$

$a^4b^7 = a \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b \cdot b \cdot b \cdot b \cdot b$

So the GCF is $a \cdot a \cdot b \cdot b = a^2b^2$

Notice that the GCF of terms containing variables will use the smallest exponent found amongst the individual terms for each variable.

Factoring Polynomials

The first step in factoring a polynomial is to find the GCF of all its terms.

Then we write the polynomial as a product by ***factoring out*** the GCF from all the terms. The remaining factors in each term will form a polynomial.

Factoring out the GCF

Example

Factor out the GCF in each of the following polynomials.

1) $6x^3 - 9x^2 + 12x =$

1) $6x - 9x + 12x =$

$3 \cdot x \cdot 2 \cdot x^2 - 3 \cdot x \cdot 3 \cdot x + 3 \cdot x \cdot 4 =$

$3x(2x^2 - 3x + 4)$

2) $14x^3y + 7x^2y - 7xy =$

$7 \cdot x \cdot y \cdot 2 \cdot x^2 + 7 \cdot x \cdot y \cdot x - 7 \cdot x \cdot y \cdot 1 =$

$7xy(2x^2 + x - 1)$

Factoring out the GCF

Example

Factor out the GCF in each of the following polynomials.

1) $6(x + 2) - y(x + 2) =$

1) $6(x + 2) - y(x + 2) =$

$6 \cdot (x + 2) - y \cdot (x + 2) =$

$(x + 2)(6 - y)$

2) $xy(y + 1) - (y + 1) =$

$xy \cdot (y + 1) - 1 \cdot (y + 1) =$

$(y + 1)(xy - 1)$

Factoring

Remember that factoring out the GCF from the terms of a polynomial should always be the first step in factoring a polynomial.

This will usually be followed by additional steps in the process.

Example

Factor $90 + 15y^2 - 18x - 3xy^2$.

$$90 + 15y^2 - 18x - 3xy^2 = 3(30 + 5y^2 - 6x - xy^2) =$$

$$3(5 \cdot 6 + 5 \cdot y^2 - 6 \cdot x - x \cdot y^2) =$$

$$3(5(6 + y^2) - x(6 + y^2)) =$$

$$3(6 + y^2)(5 - x)$$

Factoring Trinomials of the Form $x^2 + bx + c$

Factoring Trinomials

Recall by using the FOIL method that

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$$(x + 2)(x + 4) = x^2 + 4x + 2x + 8$$

$$= x^2 + 6x + 8$$

To factor $x^2 + bx + c$ into (x + one #)(x + another #),
To factor $x + bx + c$ into (x + one #)(x + another #),
note that ***b*** is the sum of the two numbers and ***c*** is
note that ***b*** is the sum of the two numbers and ***c*** is
the product of the two numbers.

So we'll be looking for 2 numbers whose product is ***c***
and whose sum is ***b***.

Note: there are fewer choices for the product, so
that's why we start there first.

Factoring Polynomials

Example

Factor the polynomial $x^2 + 13x + 30$.
Since our two numbers must have a product of 30 and

a sum of 13, the two numbers must both be positive.

Positive factors of 30	Sum of Factors
1, 30	31
2, 15	17



Note, there are other factors, but once we find a pair that works, we do not have to continue searching.

So $x^2 + 13x + 30 = (x + 3)(x + 10)$.

Factoring Polynomials

Example

Factor the polynomial $x^2 - 11x + 24$.

Since our two numbers must have a product of 24 and

a sum of -11, the two numbers must both be negative.

Negative factors of 24 Sum of Factors

– 1, – 24 – 25

– 2, – 12 – 14



So $x^2 - 11x + 24 = (x - 3)(x - 8)$.

Factoring Polynomials

Example

Factor the polynomial $x^2 - 2x - 35$.

Since our two numbers must have a product of – 35 and a sum of – 2, the two numbers will have to have different signs. Factors of – 35 Sum of Factors

Factors of – 35	Sum of Factors
– 1, 35	34
1, – 35	– 34
– 5, 7	2



So $x^2 - 2x - 35 = (x + 5)(x - 7)$.

Prime Polynomials

Example

Factor the polynomial $x^2 - 6x + 10$.

Since our two numbers must have a product of 10 and a sum of – 6, the two numbers will have to both be negative.

Negative factors of 10	Sum of Factors
– 1, – 10	– 11
– 2, – 5	– 7

Since there is not a factor pair whose sum is – 6, $x^2 - 6x + 10$ is not factorable and we call it a ***prime polynomial***.

Check Your Result!

You should always check your factoring results by multiplying the factored polynomial to verify that it is equal to the original polynomial.

Many times you can detect computational

errors or errors in the signs of your numbers by checking your results.

Factoring Trinomials of the Form $ax^2 + bx + c$

Factoring Trinomials

Returning to the FOIL method,

$$\underset{\qquad\qquad\qquad

inside coefficients.

Note that ***b*** is the sum of 2 products, not just 2 numbers, as in the last section.

Factoring Polynomials

⚠ **Example**

Factor the polynomial $25x^2 + 20x + 4$.

Possible factors of $25x^2$ are $\{x, 25x\}$ or $\{5x, 5x\}$.

Possible factors of 4 are $\{1, 4\}$ or $\{2, 2\}$.

We need to methodically try each pair of factors until we find a combination that works, or exhaust all of our possible pairs of factors.

Keep in mind that, because some of our pairs are not identical factors, we may have to exchange some pairs of

factors and make 2 attempts before we can definitely decide a particular pair of factors will not work. Continued.

Factoring Polynomials



Example Continued

We will be looking for a combination that gives the sum of the products of the outside terms and the inside terms equal to $20x$.

Factors of $25x^2$	Factors of 4	Resulting Binomials	Product of Outside Terms	Product of Inside Terms	Sum of Products
$\{x, 25x\}$	$\{1, 4\}$	$(x + 1)(25x + 4)$	$4x$	$25x$	$29x$
$\{x, 25x\}$	$\{2, 2\}$	$(x + 2)(25x + 2)$	$2x$	$50x$	$52x$
$\{5x, 5x\}$	$\{2, 2\}$	$(5x + 2)(5x + 2)$	$10x$	$10x$	$20x$

Continued.

Factoring Polynomials

⚠ **Example Continued**

Check the resulting factorization using the FOIL method.

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$(5x + 2)(5x + 2) = 5x(5x)$

$(5x + 2)(5x + 2) = \qquad 2(2)$

$5x(5x) + 5x(2) + 2(5x) +$

$+ 5x(2) \quad + 2(5x) \quad + 2(2)$

$= 25x^2 + 10x + 10x + 4$

$= 25x^2 + 20x + 4$

So our final answer when asked to factor $25x^2 + 20x + 4$ will be $(5x + 2)(5x + 2)$ or $(5x + 2)^2$.

Factoring Polynomials

⚠ **Example**

Factor the polynomial $21x^2 - 41x + 10$.

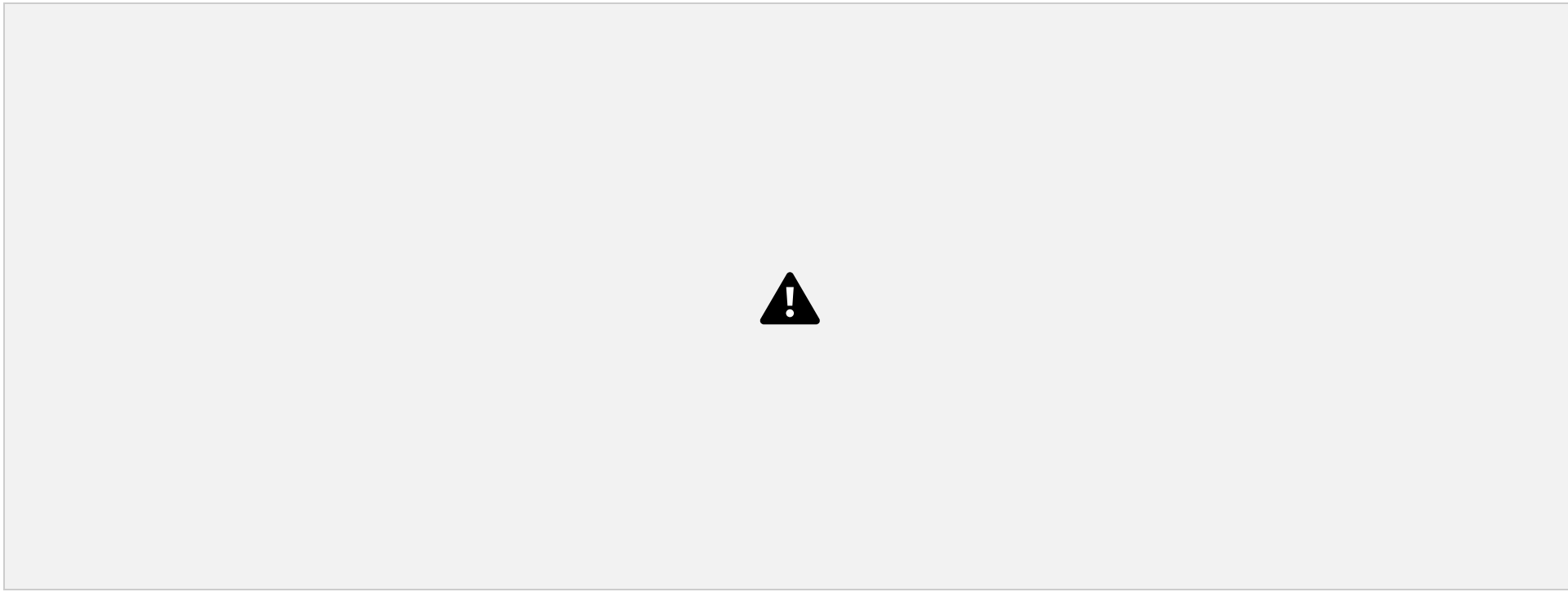
Possible factors of $21x^2$ are $\{x, 21x\}$ or $\{3x, 7x\}$.

Since the middle term is negative, possible factors of 10 must both be negative: $\{-1, -10\}$ or $\{-2, -5\}$.

We need to methodically try each pair of factors until we find a combination that works, or exhaust all of our

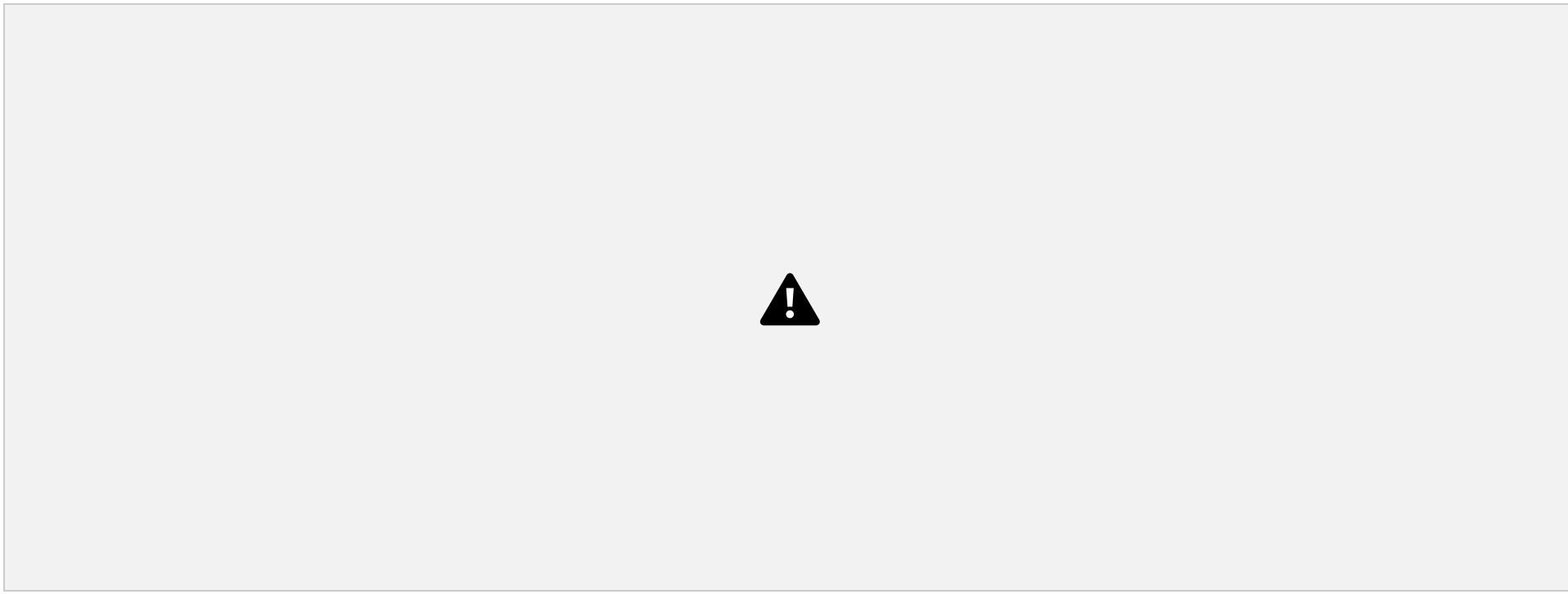
possible pairs of factors.

Continued.









Factoring Polynomials



Example Continued

We will be looking for a combination that gives the sum of the products of the outside terms and the inside terms equal to $-41x$.

Factors of Factors of Resulting Product of Outside Product of Inside Sum of

Factors of $21x^2$	Factors of 10 Resulting	Binomials Product of	Outside Terms Product of	Inside Terms Sum of	Products
$\{x, 21x\}$	$\{1, 10\}$	$(x - 1)(21x - 10)$	$-10x$	$-21x$	$-31x$
		$(x - 10)(21x - 1)$	$-x$	$-210x$	$-211x$
$\{x, 21x\}$	$\{2, 5\}$	$(x - 2)(21x - 5)$	$-5x$	$-42x$	$-47x$
		$(x - 5)(21x - 2)$	$-2x$	$-105x$	$-107x$

Continued.

Factoring

Polynomials



Example

Continued

Factors of $21x^2$ / Factors of 10	Resulting Binomials	Product of Outside Terms	Product of Inside Terms	Sum of Products
{3*x*, 7*x*}{1, 10}	(3*x* – 1)(7*x* – 10)	−30*x*	−7*x*	−37*x*
	(3*x* – 10)(7*x* – 1)	−3*x*	−70*x*	−73*x*
	(3*x* – 10)(7*x* – 1)	−3*x*	−70*x*	−73*x*
{3*x*, 7*x*} {2, 5}	(3*x* – 2)(7*x* – 5)	−15*x*	−14*x*	−29*x*
	(3*x* – 5)(7*x* – 2)	−6*x*	−35*x*	−41*x*

Continued.

Factoring Polynomials



Example Continued

Check the resulting factorization using the FOIL method.

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$(3x - 5)(7x - 2) = 3x(7x)$

$(3x - 5)(7x - 2) = \qquad 5(-2)$

$3x(7x) + 3x(-2) - 5(7x) -$

$+ 3x(-2) - 5(7x) \quad - 5(-2)$

$= 21x^2 - 6x - 35x + 10$

$= 21x^2 - 41x + 10$

So our final answer when asked to factor $21x^2 - 41x + 10$ will be $(3x - 5)(7x - 2)$.

Factoring Polynomials



Example

Factor the polynomial $3x^2 - 7x + 6$.

The only possible factors for 3 are 1 and 3, so we know that, if factorable, the polynomial will have to look like $(3x\ )(x\ )$ in factored form, so that the product of the first two terms in the binomials will be $3x^2$.
of the first two terms in the binomials will be $3x^2$.

Since the middle term is negative, possible factors of 6 must both be negative: {−1, − 6} or {− 2, − 3}.

We need to methodically try each pair of factors until we find a combination that works, or exhaust all of our possible pairs of factors.

Continued.

Factoring Polynomials

Example Continued

We will be looking for a combination that gives the sum of the products of the outside terms and the inside terms equal to $-7x$.

Factors of 6 Resulting	Binomials Product of	Factors of Resulting Product of Outside Terms	Product of Inside Terms	Sum of Products
{−1, −6}	$(3x - 1)(x - 6)$	$-18x$	$-x$	$-19x$
{−2, −3}	$(3x - 2)(x - 3)$	$-9x$	$-2x$	$-11x$

$(3x - 3)(x - 2)$ Common factor so no need to test.

$(3x - 6)(x - 1)$ Common factor so no need to test.

Continued.

⚠ Example Continued

Now we have a problem, because we have exhausted all possible choices for the factors, but have not found a pair where the sum of the products of the outside terms and the inside terms is -7.

So $3x^2 - 7x + 6$ is a prime polynomial and will not factor.

Factoring Polynomials

⚠ Example

Factor the polynomial $6x^2y^2 - 2xy^2 - 60y^2$.

Remember that the larger the coefficient, the greater the probability of having multiple pairs of factors to check. So it is important that you attempt to factor out any common factors first.

$$6x^2y^2 - 2xy^2 - 60y^2 = 2y^2(3x^2 - x - 30)$$

The only possible factors for 3 are 1 and 3, so we know that, if we can factor the polynomial further, it will have to look like $2y^2(3x\ )(x\ )$ in factored form. Continued.

Factoring Polynomials



Example Continued

Since the product of the last two terms of the

binomials will have to be –30, we know that they must be different signs.

Possible factors of –30 are { 1, 30}, {1, 30}, { 2, 15}, {2,

Possible factors of –30 are {–1, 30}, {1, –30}, {–2, 15}, {2, –15}, {–3, 10}, {3, –10}, {–5, 6} or {5, –6}.

We will be looking for a combination that gives the sum of the products of the outside terms and the inside terms equal to –*x*. Continued.

Factoring

Polynomials



Example

Continued

Factors of -30 Resulting	Binomials Product of	Outside Terms Product of	Inside Terms Sum of Products

{-1, 30} $(3x - 1)(x + 30)$ $90x$ $-x$ $89x$ $(3x + 30)(x - 1)$ Common factor so no need to test.

{1, -30} $(3x + 1)(x - 30)$ $-90x$ x $-89x$ $(3x - 30)(x + 1)$ Common factor so no need to test.

{-2, 15} $(3x - 2)(x + 15)$ $45x$ $-2x$ $43x$ $(3x + 15)(x - 2)$ Common factor so no need to test.

{2, -15} $(3x + 2)(x - 15)$ $-45x$ $2x$ $-43x$ $(3x - 15)(x + 2)$ Common factor so no need to test.

Continued.

Factoring

Polynomials

⚠ **Example**

Continued

Factors of –30 Resulting | Binomials Product of | Outside Product of | Terms Inside | Terms Sum of | Products

{–3, 10} $(3x - 3)(x + 10)$ Common factor so no need to test. $(3x + 10)(x - 3)$ $-9x$ $10x$

x

$(3x + 10)(x - 3)$ $-9x$ $10x$ x

{3, –10} $(3x + 3)(x - 10)$ Common factor so no need to test. $(3x - 10)(x + 3)$ $9x$ $-10x$



$-x$

Continued.

Factoring Polynomials

Example Continued

Check the resulting factorization using the FOIL method.

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$(3x - 10)(x + 3) = 3x(x)$

$(3x - 10)(x + 3) = \qquad 10(3)$

$3x(x) + 3x(3) - 10(x) -$

$+ 3x(3) \quad - 10(x) \quad - 10(3)$

$= 3x^2 + 9x - 10x - 30$

$= 3x^2 - x - 30$

So our final answer when asked to factor the polynomial

$6x^2y^2 - 2xy^2 - 60y^2$ will be $2y^2(3x - 10)(x + 3)$.

Factoring Trinomials of the Form $x^2 + bx + c$ by Grouping

Factoring by Grouping

Factoring polynomials often involves additional techniques after initially factoring out the GCF. One technique is ***factoring by grouping***.

⚠ Example

Factor $xy + y + 2x + 2$ by grouping.

Notice that, although 1 is the GCF for all four terms of the polynomial, the first 2 terms have a GCF of y and the last 2 terms have a GCF of 2.

$xy + y + 2x + 2 = x \cdot y + 1 \cdot y + 2 \cdot x + 2 \cdot 1 = y(x +$

$1) + 2(x + 1) = (x + 1)(y + 2)$

Factoring by Grouping

Factoring a Four-Term Polynomial by Grouping 1) Arrange the terms so that the first two terms have a common factor and the last two terms have a common factor.

2) For each pair of terms, use the distributive property to factor out the pair's greatest common factor.

3) If there is now a common binomial factor, factor it out.

4) If there is no common binomial factor in step 3, begin again, rearranging the terms differently.

- If no rearrangement leads to a common binomial factor, the polynomial cannot be factored.

Factoring by Grouping

⚠ **Example**

Factor each of the following polynomials by grouping.

1) $x^3 + 4x + x^2 + 4 = \mathbf{x} \cdot x^2 + \mathbf{x} \cdot 4 + \mathbf{1} \cdot x^2 + \mathbf{1} \cdot 4 =$

$\mathbf{x}(x^2 + 4) + \mathbf{1}(x^2 + 4) =$

$\mathbf{x}(x^2 + 4) + \mathbf{1}(x^2 + 4) =$

$(x^2 + 4)(\mathbf{x} + \mathbf{1})$

2) $2x^3 - x^2 - 10x + 5 = \mathbf{x^2} \cdot 2x - \mathbf{x^2} \cdot 1 - \mathbf{5} \cdot 2x - \mathbf{5} \cdot (-1) =$

$\mathbf{x^2}(2x - 1) - \mathbf{5}(2x - 1) =$

$(2x - 1)(\mathbf{x^2} - \mathbf{5})$

Factoring by Grouping

⚠ **Example**

Factor $2x - 9y + 18 - xy$ by grouping.

Neither pair has a common factor (other than 1). So, rearrange the order of the factors.

$2x + 18 - 9y - xy = \mathbf{2} \cdot x + \mathbf{2} \cdot 9 - 9 \cdot \mathbf{y} - x \cdot \mathbf{y} =$

$\mathbf{2}(x + 9) - \mathbf{y}(9 + x) =$

$\mathbf{2}(\mathbf{x + 9}) - \mathbf{y}(\mathbf{x + 9})$ = (make sure the factors are identical)

$(\mathbf{x + 9})(\mathbf{2} - \mathbf{y})$

Factoring Perfect Square Trinomials and the Difference of Two Squares

Perfect Square Trinomials

Recall that in our very first example in Section 4.3 we attempted to factor the polynomial $25x^2 + 20x + 4$.

The result was $(5x + 2)^2$, an example of a

The result was $(5x + 2)$

binomial squared.

Any trinomial that factors into a single binomial squared is called a ***perfect square trinomial***.

Perfect Square Trinomials

shortcut for squaring a binomial

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

So if the first and last terms of our polynomial to be factored are can be written as expressions squared, and the middle term of our polynomial is twice the product of those two expressions, then we can use these two previous equations to easily factor the polynomial. $a^2 + 2ab + b^2 = (a + b)^2$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Perfect Square Trinomials

⚠

Example

Factor the polynomial $16x^2 - 8xy + y^2$.

Since the first term, $16x^2$, can be written as $(4x)^2$, and the last term, y^2 is obviously a square, we check the

middle term.

$8xy = 2(4x)(y)$ (twice the product of the expressions that are squared to get the first and last terms of the polynomial)

Therefore $16x^2 - 8xy + y^2 = (4x - y)^2$.

Note: You can use FOIL method to verify that the factorization for the polynomial is accurate.

Difference of Two Squares

Another shortcut for factoring a trinomial is when we want to factor the difference of two squares. $a^2 - b^2 = (a + b)(a - b)$

A binomial is the difference of two square if

1. both terms are squares and

2.the signs of the terms are different.

$$9x^2 - 25y^2$$

$$-c^4 + d^4$$

Difference of Two Squares

⚠ **Example**

Factor the polynomial $x^2 - 9$.

The first term is a square and the last term, 9, can be written as 3^2. The signs of each term are different, so we have the difference of two squares

Therefore $x^2 - 9 = (x - 3)(x + 3)$.

Note: You can use FOIL method to verify that the factorization for the polynomial is accurate.

Solving Quadratic Equations by Factoring

Zero Factor Theorem

Quadratic Equations

- Can be written in the form $ax^2 + bx + c = 0$.
- a, b and c are real numbers and $a \neq 0$.
- This is referred to as ***standard form***.

Zero Factor Theorem

- If a and b are real numbers and $ab = 0$, then $a = 0$ or $b = 0$.
- This theorem is very useful in solving quadratic equations.

Solving Quadratic Equations

Steps for Solving a Quadratic Equation by Factoring

1) Write the equation in standard form.

2) Factor the quadratic completely.

3) Set each factor containing a variable equal to 0.

4) Solve the resulting equations.

5) Check each solution in the original equation.

Solving Quadratic Equations



Solve $x^2 - 5x = 24$.

- First write the quadratic equation in standard form. $x^2 - 5x - 24 = 0$
- Now we factor the quadratic using techniques from the previous sections.

  $x^2 - 5x - 24 = (x - 8)(x + 3) = 0$
- We set each factor equal to 0.

  $x - 8 = 0$ or $x + 3 = 0$, which will simplify to

  $x = 8$ or $x = -3$ Continued.

Solving Quadratic Equations

⚠ **Example Continued**

- Check both possible answers in the original equation.

$8^2 - 5(8) = 64 - 40 = 24$ ***true***

$8^2 - 5(8) = 64 - 40 = 24$ ***true***

$(-3)^2 - 5(-3) = 9 - (-15) = 24$ ***true***

- So our solutions for x are 8 or –3.

Solving Quadratic Equations

⚠ **Example**

Solve $4x(8x + 9) = 5$

- First write the quadratic equation in standard form.

$32x^2 + 36x = 5$

$32x^2 + 36x - 5 = 0$

$32x^2 + 36x - 5 = 0$

- Now we factor the quadratic using techniques from the previous sections.

$32x^2 + 36x - 5 = (8x - 1)(4x + 5) = 0$

- We set each factor equal to 0.

$8x - 1 = 0$ or $4x + 5 = 0$

$8x = 1$ or $4x = -5$, which simplifies to $x =$ or $\frac{5}{4}$. $-\frac{1}{8}$ Continued.

Solving Quadratic Equations

Example Continued

- Check both possible answers in the original equation. (

)(( )) ( )( ) ( ) 1 11 4 8 9 4 1 9 4 (1 0) (1 0) 5 8

1 1

(88)(( )) (8) (8) 2

88 2 += += = =

8

true

( )(( )) ( )( ) ( ) 55 4 8 9 4 10 9 4 (1) (5)(1) 5 4

55

4 4 4 – – + = – – + = – – =– – = *true*

- So our solutions for x are or . 1 5 8 4 –

Finding *x*-intercepts

Recall that we found the *x*-intercept of linear equations by letting *y* = 0 and solving for *x*.

The same method works for *x*-intercepts in quadratic equations.

Note: When the quadratic equation is written in

standard form, the graph is a parabola opening up (when $a > 0$) or down (when $a < 0$), where a is the coefficient of the x^2 term.

The intercepts will be where the parabola crosses the x-axis.

Finding x-intercepts



Example

Find the x-intercepts of the graph of $y = 4x^2 + 11x + 6$.

The equation is already written in standard form, so we let $y = 0$, then factor the quadratic in x. $0 = 4x^2 + 11x + 6 = (4x + 3)(x + 2)$

$$0 = 4x^2 + 11x + 6 = (4x + 3)(x + 2)$$

We set each factor equal to 0 and solve for

x. $4x + 3 = 0$ or $x + 2 = 0$

$$4x = -3 \text{ or } x = -2$$

$$x = -\tfrac{3}{4} \text{ or } x = -2$$

So the x-intercepts are the points $(-\frac{3}{4}, 0)$ and $(-2, 0)$.

Quadratic Equations and Problem Solving

Strategy for Problem Solving

General Strategy for Problem Solving 1) Understand the problem

- Read and reread the problem
- Choose a variable to represent the unknown •

Construct a drawing, whenever possible •

Propose a solution and check

2) Translate the problem into an equation

3) Solve the equation

4) Interpret the result

- Check proposed solution in problem
- State your conclusion

Finding an Unknown Number



Example

The product of two consecutive positive integers is 132. Find the two integers.

1.) *Understand*

Read and reread the problem. If we let

x = one of the unknown positive integers, then

$x + 1$ = the next consecutive positive integer.

Continued

Finding an Unknown Number



Example continued

2.) ***Translate***

The product of

The product of



is positive integers





132

two consecutive

$x\,(x+1)$ $=$ $\bullet\ 132$

Continued

Finding an Unknown Number

Example continued

3.) ***Solve***

$x(x+1) = 132$

$x^2 + x = 132$ (Distributive property)

$x + x = 132$ (Distributive property)

$x^2 + x - 132 = 0$ (Write quadratic in standard form) $(x + 12)(x -$

$11) = 0$ (Factor quadratic polynomial) $x + 12 = 0$ or $x - 11 = 0$

(Set factors equal to 0)

$x = -12$ or $x = 11$ (Solve each factor for x)

Continued

Finding an Unknown Number



Example continued

4.) *Interpret*

Check: Remember that x is suppose to represent a positive integer. So, although $x = -12$ satisfies our equation, it cannot be a solution for the problem we were presented.

If we let $x = 11$, then $x + 1 = 12$. The product of the two numbers is $11 \cdot 12 = 132$, our desired result.

State: The two positive integers are 11 and 12.

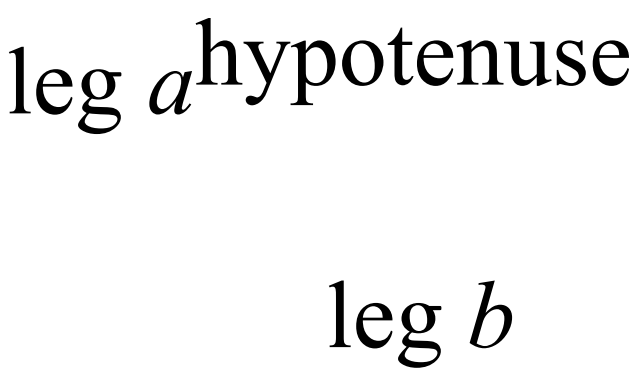
The Pythagorean Theorem

Pythagorean Theorem

In a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse.

$$(\text{leg } a)^2 + (\text{leg } b)^2 = (\text{hypotenuse})^2$$

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The Pythagorean Theorem

Example

Find the length of the shorter leg of a right triangle if the longer leg is 10 miles more than the shorter leg and the hypotenuse is 10 miles less than twice the shorter leg.

1.) *Understand*

Read and reread the problem. If we let
x = the length of the shorter leg, then

x

$x + 10$ = the length of the longer leg and
$2x - 10$ = the length of the hypotenuse.

x

$2 - 10\ x\ + 10$

Continued

The Pythagorean Theorem

⚠ **Example continued**

2.) *Translate*

By the Pythagorean Theorem,

$$(\text{leg } a)^2 + (\text{leg } b)^2 = (\text{hypotenuse})^2$$

$$x^2 + (x + 10)^2 = (2x - 10)^2$$

3.) *Solve*

$x^2 + (x + 10)^2 = (2x - 10)^2$

$x^2 + x^2 + 20x + 100 = 4x^2 - 40x + 100$ (multiply the binomials) $2x^2 + 20x + 100 = 4x^2 -$

$40x + 100$ (simplify left side) $0 = 2x^2 - 60x$ (subtract $2x^2 + 20x + 100$ from both sides)0

$= 2x(x - 30)$ (factor right side)

$x = 0$ or $x = 30$ (set each factor = 0 and solve)

Continued

The Pythagorean Theorem



Example continued

4.) ***Interpret***

Check: Remember that *x* is suppose to represent the length of the shorter side. So, although *x* = 0 satisfies our equation, it cannot be a solution for the problem we were presented.

If we let If we let *x* = 30, then = 30, then *x* + 10 = 40 and 2 + 10 = 40 and 2*x* – 10 = 50. Since 302 + 402 = 900 + 1600 = 10 = 50. Since 302 + 402 = 900 + 1600 = 2500 = 502, the Pythagorean Theorem checks out.

State: The length of the shorter leg is 30 miles. (Remember that is all we were asked for in this problem.)